

Overview of Autonomous vehicle

Subject: Autonomous vehicle

1.

Toyota In November 2022, Toyota demonstrated one of its GR Yaris test cars, which had been trained using professional rally drivers. Toyota used its collaboration with Microsoft in FIA World Rally Championship since the 2017 season.

2.

Pedestrian reactions In 2023 David R. Large, senior research fellow with the Human Factors Research Group at the University of Nottingham, disguised himself as a car seat in a study to test people's reactions to driverless cars. He said, "We wanted to explore how pedestrians would interact with a driverless car and developed this unique methodology to explore their reactions." The study found that, in the absence of someone in the driving seat, pedestrians trust certain visual prompts more than others when deciding whether to cross the road.

3.

Simulation and validation In September 2022, Biproggy released Driving Intelligence Validation Platform (DIVP) as part of Japanese national project "SIP-adus", which is interoperable with Open Simulation Interface (OSI) of ASAM.

4.

Total incidents The US National Highway Traffic Safety Administration began mandating incident reports from autonomous vehicle companies in June 2021. Some reports cite incidents from as early as August 2019, with current data available through June 17, 2024. There have been a total of 3,979 autonomous vehicle incidents (both ADS and ADAS) reported during this timeframe. 2,146 of those incidents (53.9%) involved Tesla vehicles.

5.

British definitions In British English, the word "automated" has several meanings, such as in the sentence: "Thatcham also found that the automated lane keeping systems could only meet two out of the twelve principles required to guarantee safety, going on to say they cannot, therefore, be classed as 'automated driving', preferring 'assisted driving'". The first occurrence of the "automated" word refers to a UNECE automated system, while the second refers to the British legal definition of an automated vehicle. British law interprets the meaning of "automated vehicle" based on the interpretation section related to a vehicle "driving itself" and an insured vehicle. In November 2023 the British Government introduced the Automated Vehicles Bill. It proposed definitions for related terms:

6.

Driver communications ACs complicate the need for drivers to communicate with each other, e.g., to decide which car enters an intersection first. In an AC without a driver, traditional means such as hand signals do not work (no driver, no hands).

7.

Automation levels The SAE classification of Levels is based on the role of the driver, rather than the vehicle's capabilities, although these are related in the form of a "driving mode" (aka driving scenario). The mode is determined by both an operational design domain (ODD) and a "dynamic driving requirement." The ODD is the circumstance in which the car is driving, and the driving requirement is what the system must do while remaining safe within the boundaries of that ODD. These two things define the SAE Level. Cars may therefore switch levels according to the driving mode. A human analogy might be a mode in which you can stand unassisted on one leg (the requirement) if you are on the ground (the ODD), but would need support when standing on a tightrope. Above Level 1, level differences are related to how responsibility for safe movement is divided/shared between the advanced driver-assistance system (ADAS) and driver, rather than specific driving features (see Concepts). A car might therefore have Level 3 capability for lane management (the driving requirement) up to 100 km/h (the ODD), but Level 2 at speeds faster than that. Or it might have Level 4 for driverless navigation on designated highways, but Level 2 on urban side roads. ADAS that are considered Level 1 are adaptive cruise control, emergency brake assist, automatic emergency brake assist, lane-keeping, and lane centering. ADAS that are considered Level 2 are: highway assist, autonomous obstacle avoidance, and autonomous parking. At Level 3 and above, if a car is driving and the ODD changes - for instance, a clear highway becomes a construction zone with hand-signals from a worker - the system must recognize that it is exiting its ODD and fall back accordingly, including asking for driver intervention if necessary. A fully self-driving car would therefore stay at Level 5 for all dynamic driving requirements in all circumstances (speeds, roads, terrains, visibilities, obstacle types, jurisdictions, etc.), while a car with no ADAS controlling the driving requirements would rely on the human driver for that, and stay at Level 0 regardless of the domain.

8.

Software controls the vehicle, and can provide entertainment and other services. Over-the-air updates can deliver bug fixes and additional features over the internet. Software updates are one way to accomplish recalls that in the past required a visit to a service center. In March 2021, the UNECE regulation on software update and software update management systems was published.

9.

Approaches ACs can be tested via digital simulations, in a controlled test environment, and/or on public roads. Road testing typically requires some form of permit or a commitment to adhere to acceptable operating principles. For example, New York requires a test driver to be in the vehicle, prepared to override the ADAS as necessary.

10.

In July 2020, Toyota started public demonstration rides on Lexus LS (fifth generation) based TRI-P4 with Level 4 capability. In August 2021, Toyota operated a potentially Level 4 service using e-Palette around the Tokyo 2020 Olympic Village. In September 2020, Mercedes-Benz introduced world's first commercial Level 4 Automated Valet Parking (AVP) system named Intelligent Park Pilot for its new S-Class. In November 2022, Germany's Federal Motor Transport Authority (KBA) approved the system for use at Stuttgart Airport. In September 2021, Cruise, General Motors, and Honda started a joint testing programme, using Cruise AV. In 2023, the Origin was put on indefinite hold following Cruise's loss of its operating permit. In January 2023, Holon announced an autonomous shuttle during

the 2023 Consumer Electronics Show (CES). The company claimed that the vehicle is the world's first Level 4 shuttle built to automotive standard.

11.

Path planning Path planning finds a sequence of segments that a vehicle can use to move from origin to destination. Techniques used for path planning include graph-based search and variational-based optimization techniques. Graph-based techniques can make harder decisions such as how to pass another vehicle/obstacle. Variational-based optimization techniques require more stringent restrictions on the vehicle's path to prevent collisions. The large scale path of the vehicle can be determined by using a voronoi diagram, an occupancy grid mapping, or a driving corridor algorithm. The latter allows the vehicle to locate and drive within open space that is bounded by lanes or barriers.